Tendon Routing Designs: Advantages and Disadvantages

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Helical / Cross-helical Tendon Routing

1. **Helical Tendon Routing**
   Design: 3 straight tendons + 1 helical tendon.
   Advantages: +400% workspace volume vs. pure straight routing, with only 1 extra tendon.
   Disadvantages: helical path harder to manufacture; bending–torsion coupling complicates modeling.

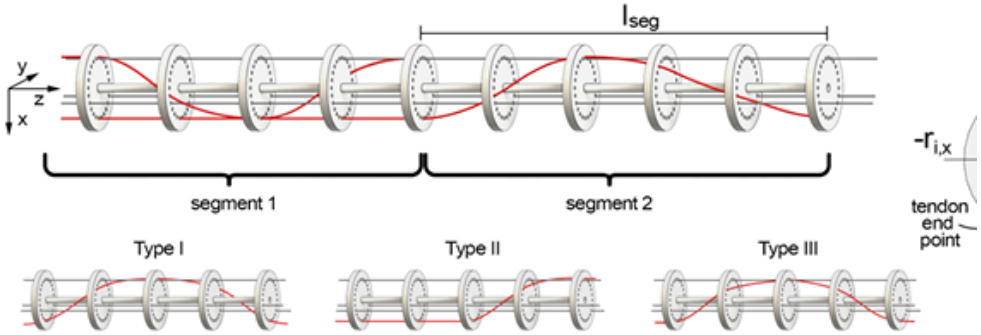


Figure 1: Helical tendon routing 1.

2. **Cross-helical Tendons**
   Design: 4 tendons, cross-helical (mirror-symmetric), single segment.
   Advantages: 1 segment produces an S-shape that normally needs 2 segments + 2 control systems.
   Disadvantages: complex, coupled force–shape relationship; harder to control precisely.

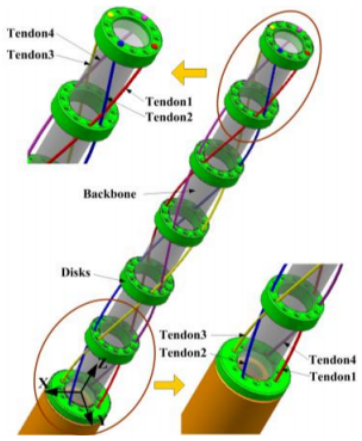


Figure 2: Helical tendon routing 2.

3. **Modular Cable-Driven Soft Arm**
   Design: 3 tendons, double-helix wound on a mandrel, then cast in silicone.

Advantages: winding + casting = 1 fabrication step; helical channel adds passive compliance for free.
Disadvantages: double-helix winding precision needed during fabrication; harder to modify after casting.

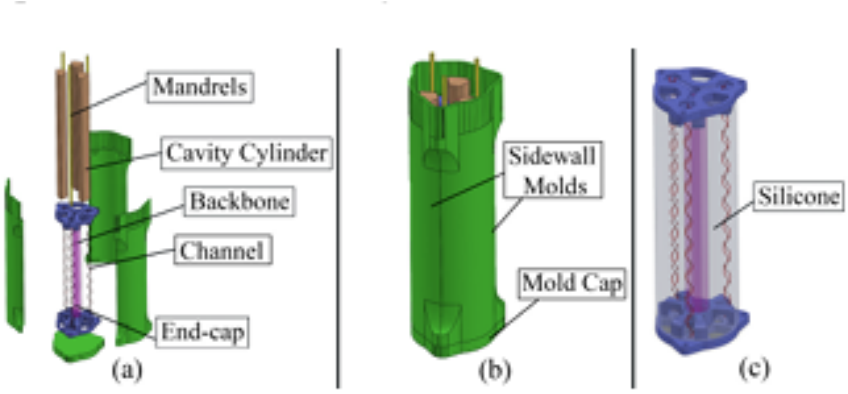


Figure 3: Helical tendon routing 3.

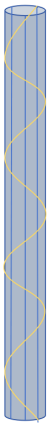


Figure 4: Helical tendon routing.

Straight / Orthogonal Tendon Routing

1. **Tapered TPU Backbone**
   Design: 3 straight tendons routed along rigid disk edges (not against the soft rod).
   Advantages: disk edges amplify the moment arm – more bending force from less material strain.
   Disadvantages: adds rigid disks, increasing weight/parts count along an otherwise soft rod.



Figure 5: Straight tendon routing 1.

2. **Compliant Spine Manipulator**
   Design: 9 tendons (3 segments × 3), straight, surface-conforming along vertebrae.
   Advantages: simple path, independent 3-segment control.
   Disadvantages: 9 tendons total – more routing/assembly effort than single-segment designs.

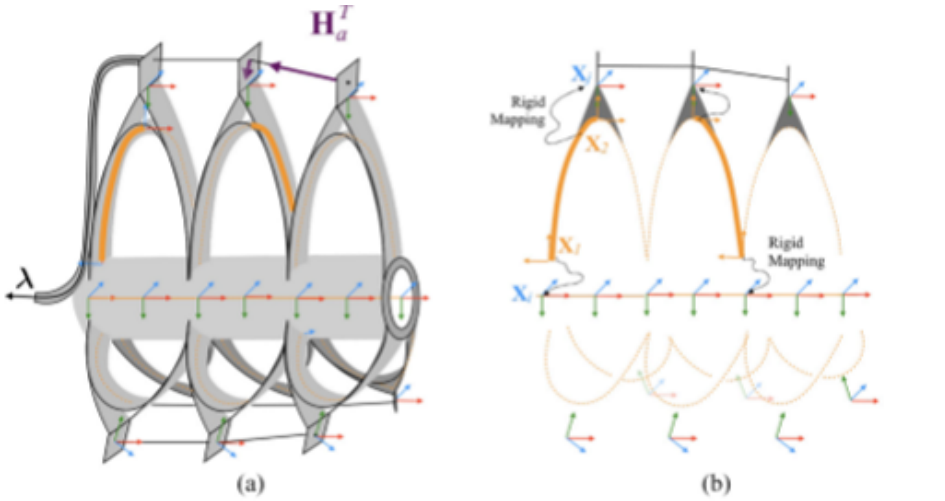


Figure 6: Straight tendon routing 5.

3. **3D Dynamics Cable-Driven Soft Manipulator**
   Design: 4 straight tendons, sheathed, only load the distal tip.
   Advantages: mid-section is passive → dynamics model simplifies to a lumped end-load problem.
   Disadvantages: sheaths add friction; can't locally control bending in the passive middle section.

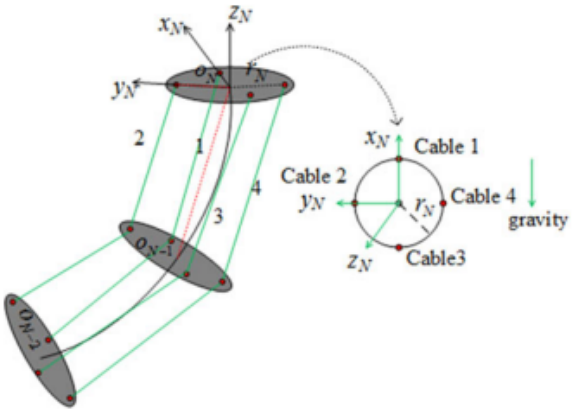


Figure 7: Straight tendon routing 6.

4. **Visual Servoing with Shape Feature**
   Design: 8 tendons, two groups anchored at different axial positions.
   Advantages: creates distinguishable local/global shape features – good observability for vision-based control.
   Disadvantages: 8 tendons is a lot of hardware/routing just to aid sensing, not to expand workspace.

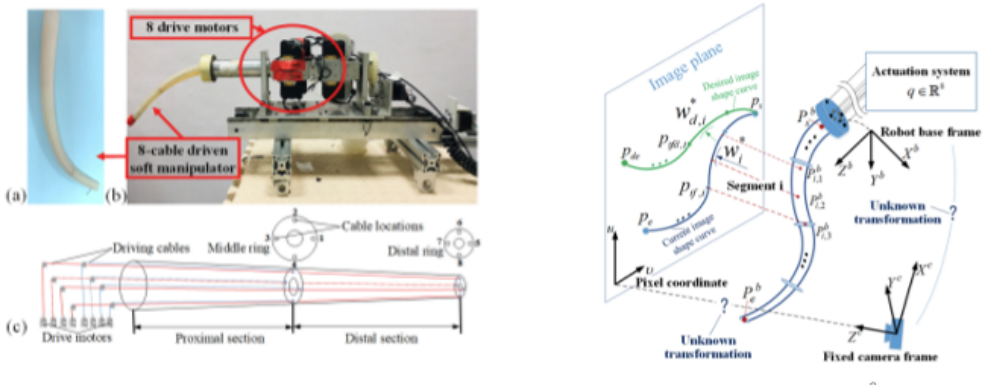


Figure 8: Straight tendon routing 7.

5. **Planar-to-Spatial Manipulator**
   Design: 4 tendons (2 per segment), two segments with orthogonal routing planes (90° offset).
   Advantages: near-complete decoupling between segments with very few tendons (only 4 total).
   Disadvantages: only 2 planes of control – less shape flexibility than fully 3D routing per segment.

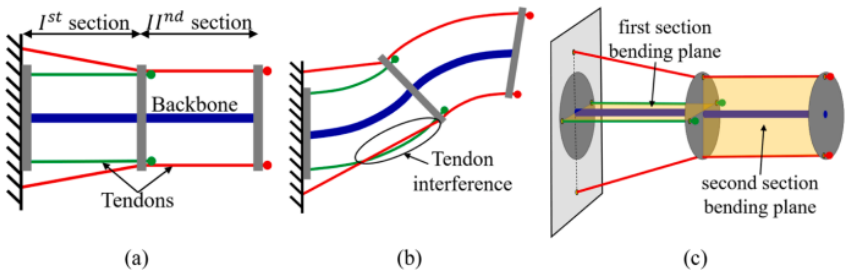


Figure 9: Straight tendon routing 8.

6. **Actuation Space Reduction (reconfigurable)**
   Design: straight tendons, but spacer disks can be actively rotated to change routing direction.
   Advantages: each disk rotation = 1 extra DOF without adding tendons.
   Disadvantages: needs active rotating hardware per disk – more actuators, more control complexity.

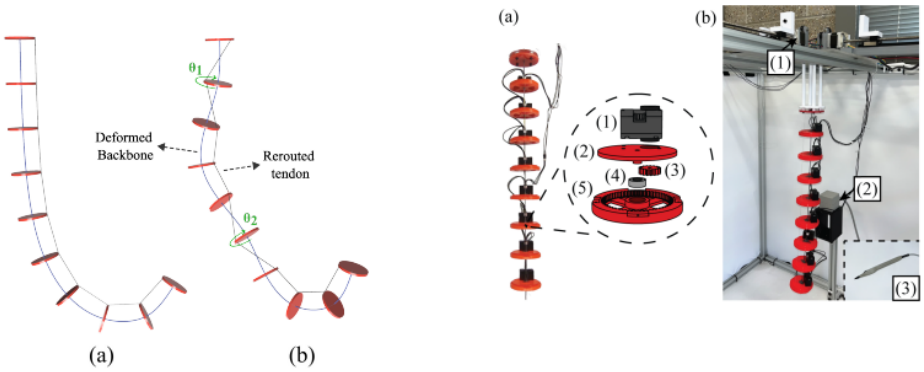


Figure 10: Straight tendon routing 9.

7. **Elephant-Trunk-Inspired Robot**
   Design: 6 straight tendons, unchanged routing; bending shaped by swapping spring stiffness per module.
   Advantages: heterogeneous curvature without touching the routing path at all – very simple tendon design.
   Disadvantages: reconfiguring stiffness (swapping springs) is a manual, discrete process, not a continuous control input.

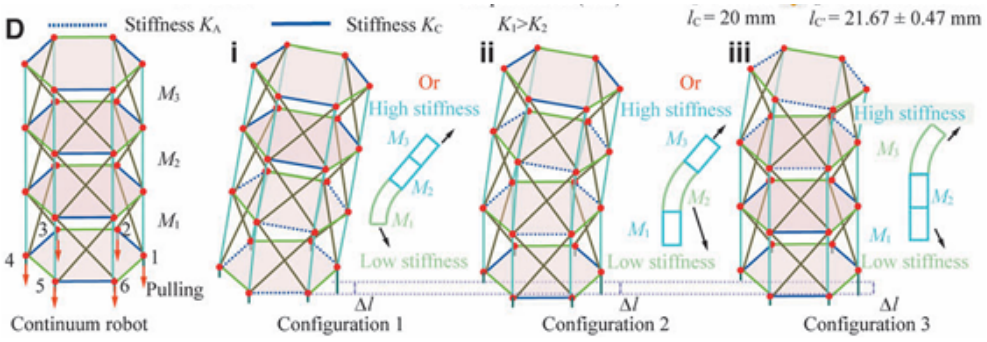


Figure 11: Straight tendon routing 10.

8. **Hybrid TDCR (torsion-avoiding)**
   Design: 1 tendon + lateral rotary joints that mechanically lock bending to one plane.
   Advantages: minimal tendon count (1!), torsion eliminated by construction, not by control.
   Disadvantages: needs a base rotation stage to redirect the bending plane – extra mechanism.

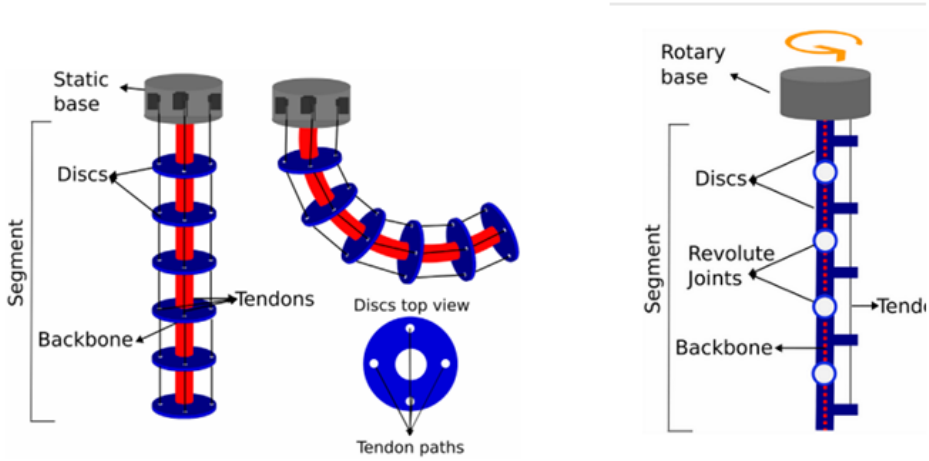


Figure 12: Straight tendon routing 11.

9. **TDCR for Medical Applications**
   Design: 4 straight tendons @ 90°, 9 springs between disks.
   Advantages: springs give equal spacing + free axial (3rd) DOF, no extra tendon needed for extension.
   Disadvantages: standard/simple design – no special workspace or decoupling advantage.

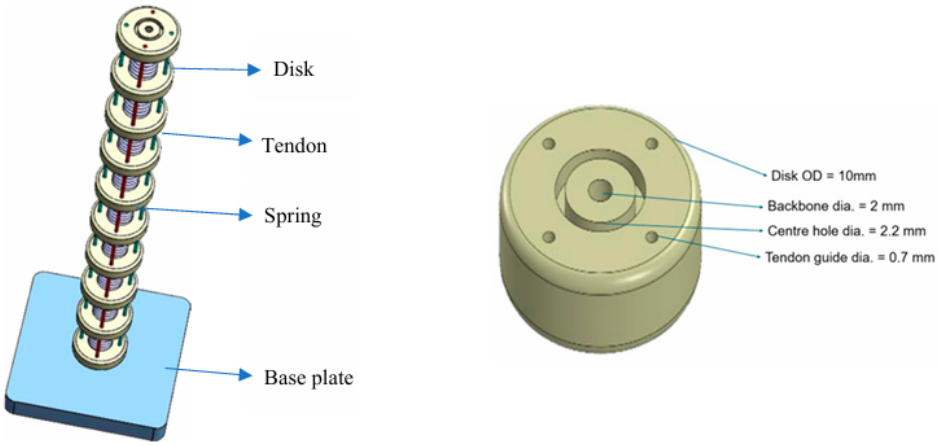


Figure 13: Straight tendon routing 12.

10. **Spring-Based Modular TDCR**
    Design: continuous spring backbone, tendons terminate at a detachable disk (proximal short-run, distal full-run).
    Advantages: effective tendon length is reconfigurable – independent control of different arc segments without changing tendon count.

Disadvantages: reconfiguring termination disks is manual/mechanical, not real-time controllable.

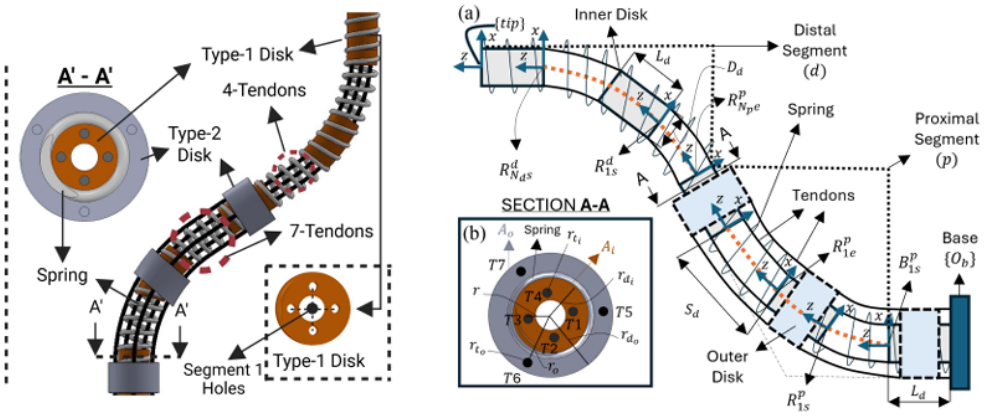


Figure 14: Straight tendon routing 13.

11. **Low-Cost TDCR (Winter-Glasgow & Borja 2024)**
Design: 3 @ 120° or 4 @ 90°, straight, off-the-shelf U-bolt guides.
Advantages: cheapest, simplest, easiest to manufacture/assemble of all surveyed designs; actuator placement decoupled from tendon exit angle via pulleys.
Disadvantages: no routing innovation – workspace/shape capability is baseline compared to helical/cross-helical designs.

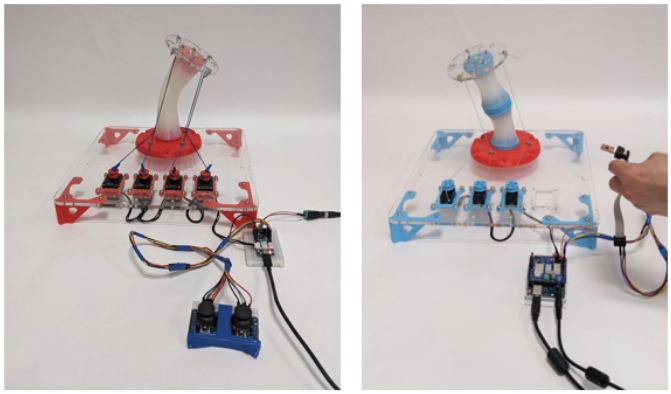


Figure 15: Straight tendon routing 14.

12. **Multi-objective Dimensional Optimization**
Design: 8 tendons (4/segment × 2), symmetric within segment, asymmetric angular offset between segments.
Advantages: offset angle becomes an optimizable variable (vs. fixed 90°) – jointly optimizes workspace and energy efficiency.
Disadvantages: 8 tendons + optimization step adds design/computation overhead vs. fixed-angle designs.

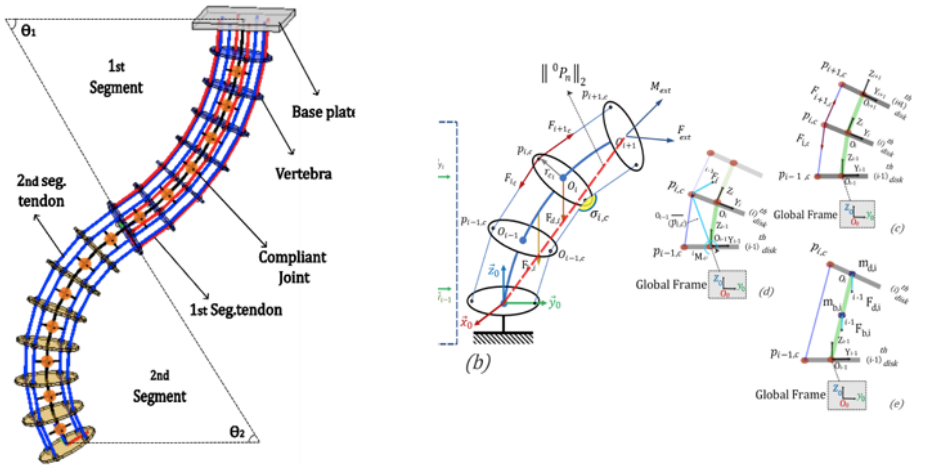


Figure 16: Straight tendon routing 15.

13. **Single-Motor Curvature-Distribution Rod**
Design: 2 tendons, 1 motor (differential drive), non-constant curvature via disk spacing d_n.
Advantages: extremely underactuated (1 motor!) yet produces non-constant curvature shapes.
Disadvantages: only bends in one plane (2 tendons); curvature profile is fixed by disk spacing, not freely reprogrammable.

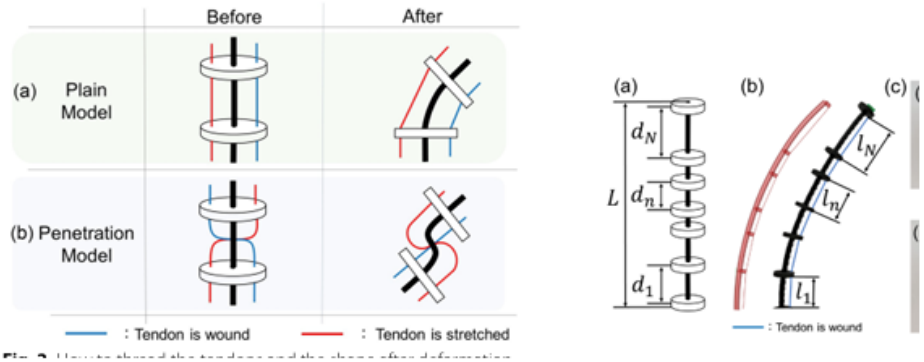


Figure 17: Straight tendon routing 16.

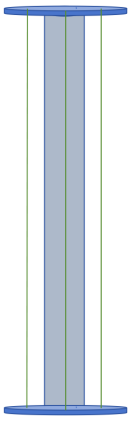


Figure 18: Straight tendon routing.

Body Geometry Is Helical, but Tendon Routing Stays Straight

1. **Trimmed Helicoids (TH)**
   Design: body itself is a trimmed helicoid; tendons stay straight, simple triangular layout.
   Advantages: large workspace, high precision, compliant interaction – all from body geometry, routing stays simple.
   Disadvantages: body fabrication (helicoid structure) is more complex than the routing itself.

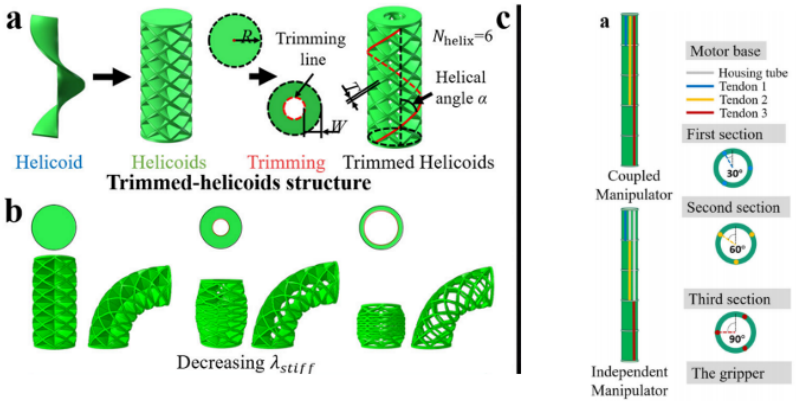


Figure 19: Helical body geometry 1.

2. **SpiRobs**
   Design: body is a log-spiral; tendons stay straight (2-cable or 3-cable).
   Advantages: 2-cable is very simple (antagonistic curl); 3-cable adds full grasping with zero routing complexity.
   Disadvantages: behavior is tied to the fixed spiral body shape – less tunable than active routing designs.

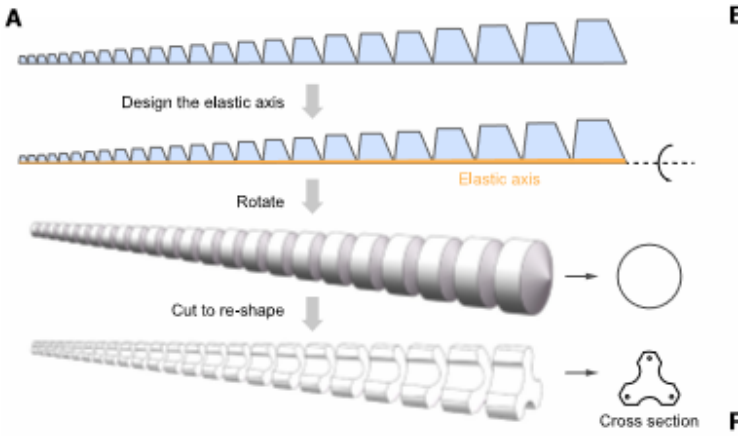


Figure 20: Helical body geometry 2.

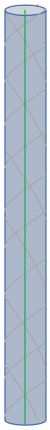


Figure 21: Helical body geometry.

Implicit / Internal-Channel Routing

1. **Hollow TDCR (reference framework)**
   Design: implicit internal routing – path emerges from tube deformation, not pre-fixed.
   Advantages: path becomes a continuous/differentiable design variable, not a discrete choice.
   Disadvantages: no fixed channel means more complex modeling; harder to predict/manufacture a specific path a priori.

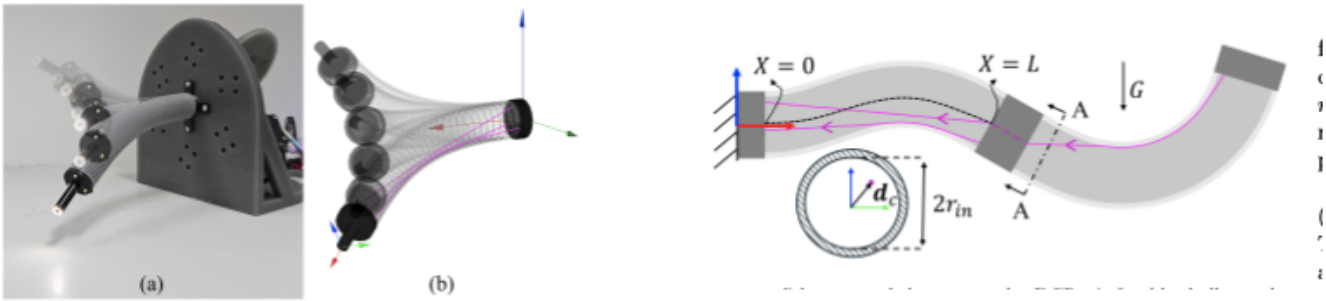


Figure 22: Internal-Channel geometry 1.

2. **Internal Tendon Routing Channels**
   Design: pre-defined internal channels inside a solid cylinder, tendon touches only the endpoints.
   Advantages: no side-contact friction – clean force transmission, simpler modeling.
   Disadvantages: channels must be manufactured internally (harder than external grooves/holes).

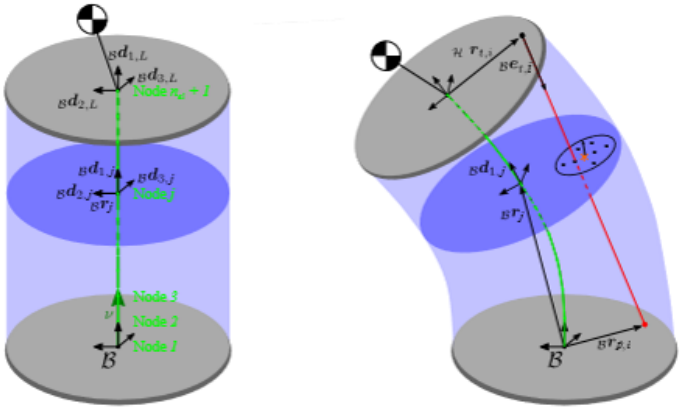


Figure 23: Internal-Channel geometry 2.

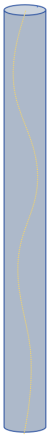


Figure 24: Internal tendon routing.